

Task 7 - Responsible and Sustainable AI

Bias, Fairness, Governance and Sustainability Report

Scope. This report evaluates responsible-AI considerations for the capstone mobility solution. The analysis uses the cleaned traffic dataset developed in Part 2 and the proxy high-risk classification target defined in Part 3. No real accident dataset was supplied, so the classification target must not be interpreted as observed accident occurrence or as a real accident predictor.

1. Bias and Fairness

Sampling and coverage limitations. The dataset contains unequal numbers of observations across years, so annual comparisons can partly reflect different data coverage rather than changes in traffic demand alone. Part 1 identified 2,559 observations in 2012, 8,573 in 2013, 4,839 in 2014, 4,373 in 2015, 9,306 in 2016, and 10,605 in 2017. Coverage is also uneven across weather, holidays and time periods. This means model performance may be less reliable for relatively sparse conditions and should not automatically be assumed to generalise to other corridors, years or locations.

Proxy-label limitations. The high_risk target is constructed from the interaction of High/Severe congestion with severe or low-visibility weather. It is a rule-based proxy created because no accident records were supplied. It measures the presence of traffic and weather conditions defined as risky by the assignment, not whether an accident actually occurred. The high classification scores therefore demonstrate that the models can reproduce this rule-based target, but they do not establish real accident-prediction performance. The target definition may also encode assumptions about what constitutes risk.

Uneven error risk. Prediction errors may not be distributed evenly across time and operating conditions. Errors can plausibly be higher in under-represented periods, unusual weather, holidays, or combinations of conditions with fewer training examples. The project therefore treats overall accuracy and related metrics as insufficient on their own for operational deployment. Before real use, performance should be checked by time period, weather condition, day type and other relevant operating groups, with subgroup sample sizes reported alongside the metrics.

2. Governance and Oversight

Human oversight. The models should be treated as decision-support tools rather than autonomous decision makers. Before real-world use, a mobility team should review data provenance, target definitions, model performance, drift alerts and subgroup performance. Model versions and changes should be documented, approval should be required before production release, and monitoring thresholds should be reviewed against operational evidence rather than treated as fixed by this academic implementation.

Deployment controls. The Part 3 prototype uses MLflow for experiment tracking, model version metadata, a FastAPI deployment mock-up and drift monitoring. These controls support traceability, but a production system would also require access control, model rollback procedures, documented ownership, incident handling and periodic reassessment. Monitoring alerts should trigger investigation rather than automatic operational action.

3. Sustainability and Resource Trade-offs

Computational choices. The project uses a mixture of linear, tree-based and neural-network models. Simpler linear models are inexpensive to train and serve, while ensemble and neural-network approaches can improve predictive performance at higher computational cost. The final system should therefore balance predictive benefit against training time, memory use, inference cost and the frequency with which models need to be retrained.

Practical sustainability. For this corridor-level use case, retraining only when monitoring indicates meaningful drift can avoid unnecessary computation. Model artifacts and experiments are tracked so work can be reproduced rather than repeatedly rebuilt without a record. Where a simpler model provides sufficiently useful performance, its lower resource requirements may be preferable to a more complex model with only marginal improvement.

4. Overall Assessment

The capstone demonstrates responsible-AI practices at an academic prototype level, but important limitations remain. The strongest limitation is the absence of real accident outcomes and the resulting proxy target. Unequal data coverage also limits direct generalisation. For real-world mobility decisions, the solution would require validated operational data, subgroup performance testing, human oversight, model governance, drift monitoring and documented approval processes. The prototype should therefore be used to demonstrate an analytical workflow, not as a deployed accident-risk decision system.